The Harmonic Architecture of the Solar System

A Silver-Ratio-Based Hypothesis for Planetary Spacing

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Code & data: github.com/salahealer9/harmonic-architecture-solar-system

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Corrected numerical reporting and declared limitations; SRHF mathematical prescription unchanged.

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Abstract

We examine whether planetary semi-major axes can be described by a phenomenological spacing prescription based on the silver ratio $\delta_S = 1 + \sqrt{2}$. The Silver-Ratio Harmonic Framework (SRHF) generates an algebraic sequence from Mercury to Pluto, including a hypothetical, unobserved node near 2.14 AU between Mars and Jupiter (*Harmonia*).

For nine observed bodies (eight planets and Pluto, a dwarf planet), excluding Harmonia, the fixed sequence gives MAPE 0.792970% and RMSE 0.105821 AU using the stated rounded orbital inputs and Earth = 1 AU normalisation. On the same six classical planets, SRHF, Titius–Bode and idealised Kepler solids yield MAPE 0.8954%, 2.6806% and 17.5152%, respectively, under explicit prescriptions.

Fitting the five outer bodies separately gives $a_H^{\mathrm{opt}} = 2.143687394$ AU, close to the fixed silver node $a_H^{\mathrm{Silver}} = 2.142135624$ AU and the transcendental reference $a_H^{\pi} = \pi^{2/3} = 2.145029397$ AU. Their full silver-to-π separation is approximately 0.1351%. The Harmonic Spread Index measures residual dispersion, not demonstrated dynamical stability. Physical interpretation and statistical rarity remain open questions.

Keywords: planetary architecture; harmonic ratios; Solar System formation; orbital resonances; silver ratio; semi-major axes; phenomenological models; exoplanetary systems.

1. Introduction

The distribution of the semi-major axes of the Solar System shows approximate regularities that have inspired many analytic schemes, from the Titius–Bode relation (Nieto, 1972) to the geometrical constructions of early Kepler (Kepler, 1619). While the formation of the modern planetary population has been explained mainly by accretion and migration (Paardekooper and Johansen, 2018), analytic patterns remain of interest as testable phenomenological descriptions. Long-term dynamical studies show that stable resonant structures and chaotic zones coexist (Laskar, 1990); such dynamics provide context, but do not independently establish the physical origin of the SRHF spacing prescription.

Here we examine a compact algebraic model based on the silver ratio $\delta_S = 1 + \sqrt{2}$. Motivated by the SRHF geometric illustration, we specify a planetary-spacing sequence through ratios and algebraic combinations of constants in $\mathbb{Q}(\sqrt{2})$. On the adopted nine-body sample, the resulting model distances have sub-per-cent mean absolute percentage error (some individual residuals exceed 1%). The focus of this study is not cosmological mechanisms but quantitative description and testable prediction.

We evaluate the model by:

1. specifying and evaluating a geometry-motivated algebraic planetary-spacing sequence;
2. computing deviations against observed semi-major axes;
3. comparing performance with existing analytic models;
4. minimising the RMSE to locate the optimal harmonic node;
5. providing falsifiable observational predictions.

In this model, a location near $a \simeq 2.14$ AU between Mars and Jupiter serves as a natural harmonic boundary, dividing the inner and outer Solar System. For brevity we refer to this location as the *Harmonia* node.

Principle 1 (Silver-Ratio Harmonic Ansatz)**.** *As a phenomenological ansatz, selected orbital radii in a planetary system are* represented *by rational functions of the silver ratio $\delta_S = 1+\sqrt{2}$ relative to a reference orbit. This parameterisation is not itself a proof of dynamical stability. In the Solar System realisation considered here, the critical resonance-like boundary is marked by the harmonic ratio*

$$\frac{2(\delta_S - 1)(2\delta_S - 3)}{\delta_S} \approx 2.142 \text{ AU}.$$

2. Harmonic Hypothesis and Mathematical Framework

2.1. Statement of the Silver-Ratio Harmonic Hypothesis

Principle 2 (Silver-Ratio Harmonic Hypothesis)**.** *We investigate the hypothesis that selected planetary semi-major axes can be represented by rational functions of the silver ratio $\delta_S = 1 + \sqrt{2}$, scaled to a reference orbit. For the Solar System, the proposed algebraic division between the inner and outer portions of the model is located at the ratio*

$$a_{\mathrm{H}} = a_{\oplus} \cdot \frac{2(\delta_S - 1)(2\delta_S - 3)}{\delta_S} \approx 2.142 \text{ AU}, \tag{1}$$

which defines the Harmonia *node between Mars and Jupiter.*

This work does not propose a new force of nature. Instead, it examines whether a simple method based on the silver ratio can accurately describe the spacing of planets, while also generating testable predictions.

2.2. SRHF Constants and Silver-Ratio Formulation

Let

$$A = \sqrt{2}, \tag{2}$$
$$B = 1 + A = 1 + \sqrt{2}, \tag{3}$$
$$C = 2A - 1 = 2\sqrt{2} - 1, \tag{4}$$
$$D = A - 1 = \sqrt{2} - 1, \tag{5}$$

so that (A, B, C, D) generate the quadratic field $\mathbb{Q}(\sqrt{2})$ forming the basis of the harmonic sequence. Figure 1 illustrates the selected constants geometrically; the planetary-spacing coefficients are specified through their algebraic ratios and combinations, rather than deduced from the diagram.

Introducing the silver ratio $\delta_S = 1 + \sqrt{2}$, we can rewrite

$$\delta_S = 1 + \sqrt{2}, \tag{6}$$
$$A = \delta_S - 1, \tag{7}$$
$$B = \delta_S, \tag{8}$$
$$C = 2\delta_S - 3, \tag{9}$$
$$D = \delta_S - 2, \tag{10}$$

so that all orbital ratios are expressed purely in terms of δ_S.

2.3. The Complete Sequence in Silver Ratio Terms

Rewriting the harmonic sequence purely in terms of δ_S:

$$a_{\text{Mercury}} = a_\oplus \cdot \tfrac{14}{15}(\delta_S - 2) \tag{11}$$
$$a_{\text{Venus}} = a_\oplus \cdot \tfrac{1}{\delta_S - 1} \tag{12}$$
$$a_{\text{Earth}} = a_\oplus \cdot 1 \tag{13}$$
$$a_{\text{Mars}} = a_\oplus \cdot \tfrac{2\delta_S - 2}{2\delta_S - 3} \tag{14}$$
$$a_{\text{Harmonia}} = a_\oplus \cdot \tfrac{2(\delta_S - 1)(2\delta_S - 3)}{\delta_S} \tag{15}$$
$$a_{\text{Jupiter}} = a_\oplus \cdot 2(\delta_S - 1)(2\delta_S - 3) \tag{16}$$
$$a_{\text{Saturn}} = a_\oplus \cdot 2(\delta_S - 1)(2\delta_S - 3)^2 \tag{17}$$
$$a_{\text{Uranus}} = a_\oplus \cdot 4(\delta_S - 1)(2\delta_S - 3)^2 \tag{18}$$
$$a_{\text{Neptune}} = a_\oplus \cdot 2(\delta_S - 1)(2\delta_S - 3)\, \tfrac{\delta_S}{\delta_S - 2} \tag{19}$$
$$a_{\text{Pluto}} = a_\oplus \cdot 4(\delta_S - 1)(2\delta_S - 3)(2\delta_S - 1) \tag{20}$$

2.4. The Harmonia Gap

The distinguished semi-major axis

$$a_{\text{H}} = a_\oplus \cdot \frac{2(\delta_S - 1)(2\delta_S - 3)}{\delta_S} \approx 2.142136 \text{ AU} \tag{21}$$

lies between Mars and Jupiter near the inner edge of the main-belt region; its role as a boundary belongs to the model and is not evidence of a detected planet or measured resonance. Figure 2 illustrates its location.

3. Empirical Fit and Residual Diagnostics

In this section we quantify the agreement between the fixed SRHF sequence and the adopted planetary semi-major axes, then examine how the five-body outer-system fit changes as the trial Harmonia parameter a_H varies. The term "basin" refers only to the descriptive shape of the RMSE curve, not to a gravitational potential or a demonstrated dynamically stable region.

3.1. Optimisation of the Harmonia Node

Figure 3a plots the RMSE of the five outer-body predictions against a_H on $[2.12, 2.18]$ AU. The fitted least-squares minimum is

$$a_H^{\text{opt}} = 2.143687394351 \text{ AU}. \tag{22}$$

The independently defined algebraic Silver node is $a_H^{\text{Silver}} = 2.142135623731$ AU; the transcendental reference is $a_H^{\pi} = \pi^{2/3} = 2.145029397111$ AU. Relative to the fitted optimum, the lower and upper separations are approximately 0.0724% and 0.0626%, respectively. The full

Silver-to-π interval is approximately 0.1351% relative to the lower node. These references are close but not coincident, and the fitted optimum must not be counted as an independent prediction.

3.2. RMSE Landscape and Harmonic Spread Index

Figure 3b presents the outer-five RMSE (indigo) and the Harmonic Spread Index (HSI; gold). HSI is the population standard deviation of five percentage residuals for the same trial a_H. Since the predicted outer semi-major axes have the form $c_i a_H$, the frozen definition simplifies for $a_H > 0$ to

$$\mathrm{HSI}(a_H) = 100\, a_H\ \mathrm{std}_i\left(\frac{c_i}{a_{i,\mathrm{obs}}}\right) = 0.296197294845\, a_H, \tag{23}$$

with HSI expressed in percentage points when a_H is in AU. Its curve is increasing and has no interior minimum around a_H^{opt}. The RMSE minimum describes fitting sensitivity, not orbital dynamics. It does not establish a gravitational attractor or physical stability basin.

3.3. Residual Distribution by Planet

Figure 4a compares signed percentage residuals for the fixed Silver node, the empirically fitted outer-five optimum, and the $\pi^{2/3}$ reference. Figure 4b is a grouped residual bar chart, with the three candidate HSI values reported in its legend; it is *not* a separate HSI-versus-a_H minimum. Small changes in these residual patterns do not independently demonstrate a stable orbital configuration.

3.4. Logarithmic Residual Structure and Neptune Zero Crossing

Figure 5a displays the outer-five RMSE/HSI diagnostics and absolute per-body residual magnitudes on a logarithmic axis, while Figure 5b shows signed residuals on a symmetric-logarithmic axis. These are alternative visual transformations of the same descriptive fit calculations.

Neptune's signed residual vanishes at $a_H = 2.143120935$ AU; the five-body RMSE is minimised separately at $a_H = 2.143687394$ AU. These different locations prevent treating Neptune's sign change as an independent marker of the global optimum. Neither a residual zero nor its logarithmic display establishes physical resonance.

3.5. Ratios, Percent Deviations, and Log-Scale Validation

Figure 6a compares *semi-major axes in AU*, not ratios, for the full sequence. The hypothetical Harmonia position is plotted as a prediction only. Figure 6b separately plots observed/predicted ratios for six observed bodies from Mars through Pluto, using explicitly distinguished fixed and fitted variants; Harmonia is not an observed entry in that ratio panel.

Figure 7a plots the nine observed-body percentage residuals, with a separate theoretical comparison of the three Harmonia reference positions. The nine-body MAPE is 0.792970%. A mean absolute percentage error describes an average absolute deviation and is *not* a confidence

band, prediction interval, or bound on every residual. The three Harmonia references span approximately 0.1351% from Silver to π (Silver denominator).

Figure 7b shows the nine measured reference values and the corresponding fixed SRHF predictions using a logarithmic ordinate; its observational score excludes Harmonia. The Mercury-to-Pluto semi-major-axis interval spans about two decades, not four.

3.6. Planet-by-Planet Comparison

Table 1 lists the adopted rounded observations and fixed algebraic predictions. The theoretical Harmonia node is displayed as an unobserved model location and is excluded from both numerical scores.

4. Comparative Model Performance

4.1. Comparison with Other Models

A quantitative comparison requires a common observational sample, identical error definitions and an explicit specification of every model's assumptions. We therefore report two different protocols in Table 2. For C6, all models are evaluated on the same six classical planets (Mercury, Venus, Earth, Mars, Jupiter, Saturn), with Earth as the 1 AU scale anchor. The fixed SRHF prescription has MAPE 0.8954% and RMSE 0.04553 AU; the specified Titius–Bode sequence gives 2.6806% and 0.18300 AU; and the idealised Kepler-solids prescription gives 17.5152% and 1.36891 AU.

For C9, the fixed SRHF is evaluated on nine observed bodies, including Pluto (a dwarf planet), but excluding the unobserved Harmonia node. The corresponding MAPE and RMSE are 0.792970% and 0.105821 AU. The C9 Titius–Bode comparison is an explicitly *conditional ordered extrapolation*, with MAPE 15.8270% and RMSE 12.9002 AU. The historical Kepler prescription is not extrapolated to C9. These comparison results establish relative fit under the specified protocols, not uniqueness, dynamical stability or a chance probability.

The distinct historical comparison of *adjacent-radius ratios* is dimensionless. Its previously reported Kepler RMSE near 0.24 has no AU units and cannot be used as the Kepler RMSE entry in a semi-major-axis table. The original historical Celtic ratio sequence is likewise not identical to adjacent ratios implied by the fixed SRHF radii.

4.2. Statistical Significance

The descriptive agreement of the fixed SRHF sequence with the adopted nine-body dataset does not by itself supply a valid chance probability or p-value. The historical heuristic multiplied a nominal tolerance probability across ten slots,

$$Q_{\text{heuristic}} = (0.04)^{10} \approx 1.05 \times 10^{-14}, \tag{24}$$

but this expression is *not* an established probability for the observed Solar System. In particular, the 0.04 per-slot value is not derived from a specified null distribution; the orbital locations cannot be assumed independent; Harmonia is not observed; and the role of model

choice and any fitting must be accounted for. The number therefore must not be interpreted as a probability, upper bound or proof of statistical significance. Establishing statistical rarity would require a separate, independently specified null-model investigation that applies the same selection and fitting procedure to comparison systems. No such test is claimed here.

4.3. Interpretation

The fixed SRHF sequence gives a compact phenomenological description of the adopted nine-body Solar System dataset, with MAPE 0.792970% and RMSE 0.105821 AU. Under the declared common-sample C6 protocol it also fits the six classical planetary radii more closely, by the chosen descriptive errors, than the specified Titius–Bode and ideal Kepler constructions. These comparisons do not, on their own, establish that the Solar System is physically quantised, that the fitted pattern is statistically rare, or that the selected algebraic construction is unique among competing spacing formulas.

4.4. Theoretical Context: Geometric Parameterisation and Fit Sensitivity

The SRHF constants (A, B, C, D) are functions of $\sqrt{2}$ motivated by the stated circle-and-grid geometry. Inserting these constants into the prescribed sequence defines the modelled orbital radii. The agreement with the observations is evaluated *after* specifying that sequence; it is not a derivation of planetary equilibrium from Newtonian gravity.

For the five outer bodies, the least-squares diagnostic is

$$\mathcal{E}(a_H) = \sqrt{\frac{1}{5}\sum_{i=1}^{5}(c_i a_H - a_{i,\mathrm{obs}})^2}, \tag{25}$$

whose minimum occurs at $a_H^{\mathrm{opt}} = 2.143687394351$ AU. This is a property of a one-parameter *fit*, while the algebraic Silver node a_H^{Silver} and transcendental reference a_H^{π} are separately defined values. Their numerical proximity is a descriptive feature worthy of further investigation, not a proved physical attractor or dynamical stability condition. Establishing orbital longevity, resonant structure, or any proposed formation mechanism would require separate gravitational and observational studies; none is claimed here.

4.5. Methods Summary

The numerical description uses the fixed algebraic constants $A = \sqrt{2}$, $B = 1+\sqrt{2}$, $C = 2\sqrt{2}-1$ and $D = \sqrt{2}-1$. The adopted observational inputs are the rounded semi-major axes listed in Table 1. Earth is both the 1 AU model-normalisation anchor and a member of each reported observational score. Pluto is a dwarf planet; Harmonia is a hypothetical, unobserved location, and is never a data point in the observational metrics. The rounded inputs constitute an author-declared compiled reference baseline, not a documented single-query extraction from JPL Horizons; their individual source editions and reference epochs have not been fully reconstructed.

With N observed bodies, the descriptive errors are

$$\mathrm{MAPE} = \frac{100}{N}\sum_{i=1}^{N}\left|\frac{a_{i,\mathrm{model}} - a_{i,\mathrm{obs}}}{a_{i,\mathrm{obs}}}\right|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(a_{i,\mathrm{model}} - a_{i,\mathrm{obs}})^2}.$$

The fixed Silver model is scored without fitting the Harmonia parameter. Separately, the empirical optimum is computed only on Jupiter, Saturn, Uranus, Neptune and Pluto, using $a_{i,\mathrm{model}}(a_H) = c_i a_H$ and least squares,

$$a_H^{\mathrm{opt}} = \frac{\sum_i c_i a_{i,\mathrm{obs}}}{\sum_i c_i^2} = 2.143687394351 \text{ AU}.$$

This fitted result is not an independent prediction. The residual-spread diagnostic is $\mathrm{HSI}(a_H) = \mathrm{std}_i(100(c_i a_H - a_{i,\mathrm{obs}})/a_{i,\mathrm{obs}})$ with population standard deviation for the same five outer bodies. For $a_H > 0$, $\mathrm{HSI}(a_H) = 0.296197294845\, a_H$, with HSI in percentage points when a_H is in AU. It has no interior minimum and is not a dynamical stability indicator.

The reproducible analytical and numerical comparisons are documented in the SRHF-A0 audit and its Python tests. Eleven plotted manuscript figure assets have identified generator scripts; three supplied figure assets do not have independently matched generators. All figures are therefore not claimed to have identical script provenance.

5. Predictions and Falsifiability

5.1. Falsifiability and Observational Tests

The fixed SRHF formula singles out an algebraic location $a_H^{\mathrm{Silver}} = 2.142135624$ AU between Mars and Jupiter. This is a model prediction for a *location*, not a prediction that an additional planet has already been observed there. The nearby $a_H^{\mathrm{opt}} = 2.143687394$ AU is an empirically fitted outer-system diagnostic, so it must not be used as an independently predicted test target.

One possible future test would examine whether a pre-specified statistic of small-body orbital elements shows an unusual feature near the fixed location. A testable analysis would need an independently chosen interval width, an explicit background or dynamical comparator, selection corrections, and uncertainty treatment. Neither such a population analysis nor a detection of a predicted dynamical resonance is established in the present work. The absence of a planet at the algebraic node does not alone falsify a spacing model that includes an unoccupied theoretical slot.

5.2. Exoplanet Systems

A possible generalisation to other planetary systems would take the ansatz

$$a_n^{\mathrm{system}} = a_{\mathrm{ref}}^{\mathrm{system}}\, R_n(\delta_S). \tag{26}$$

Whether a specified mapping and comparison protocol can describe multiple independent exoplanet systems is untested here. A future evaluation would require a predeclared planet-

selection rule, scale reference and comparator models; it is not part of the v3.1 correction.

5.3. Harmonic Ladder Structure

Figure 8 illustrates the sequence of discrete algebraic entries derived from the Silver Ratio $\delta_S = 1 + \sqrt{2}$. "Levels" and "harmonic ladder" describe the model's mathematical indexing. They do not mean that planetary orbits have experimentally established quantum energy levels or that a microscopic quantisation mechanism has been demonstrated.

6. Discussion and Limitations

The prescribed SRHF model reproduces the adopted nine observed-body semi-major axes with MAPE 0.792970% and RMSE 0.105821 AU (Earth included as the 1 AU normalisation anchor; Pluto included as a dwarf planet). The separate outer-five optimum and the close Silver/π reference values characterise its fitting geometry. They do not prove a gravitational equilibrium, a planet at Harmonia, or a physical mechanism linking δ_S to planetary formation.

Several limitations remain relevant:

- **Observational sample and data provenance.** The model is compared with nine adopted rounded semi-major axes; Harmonia is unobserved. Orbital eccentricity, inclination, mass and small-body distributions are not modelled. The adopted values form an author-declared compilation of legacy NASA reference materials, standard literature and published planetary listings, rather than outputs of one documented JPL Horizons query. Their individual source editions and orbital-element conventions have not been fully reconstructed.
- **Phenomenological, not dynamical.** No integration of an N-body system, resonance-width calculation, perturbation analysis or demonstration of long-term stability is supplied. The RMSE minimum measures fit quality and HSI measures residual spread only.
- **No established significance probability.** The historical $(0.04)^{10}$ product is an unvalidated heuristic rather than a p-value or upper bound; its factors, independence assumptions and treatment of model selection have not been justified under a declared null distribution. The question remains open without asserting that the observed agreement is insignificant.
- **Model selection and comparison scope.** The model is an algebraic construction assessed on the same Solar System data that motivated the hypothesis. C6 and C9 are precisely defined descriptive benchmarks, not exhaustive comparisons against all possible functions, population models or fitted alternatives.
- **Figure and source reproducibility.** Eleven figure assets have identifiable script generators in the historical repository; three additional figures were supplied as graphics without matched generator scripts. Their provenance is recorded without assuming full source-level reproduction.

The possibility that the spacing pattern reflects formation dynamics, resonance capture, or an effective organising principle remains a legitimate topic for future research, but no such

causal conclusion is required or established by the results reported here.

7. Speculative Connections Across Scales

7.1. The Bohr–Solar Correspondence

Definition 1. ***Quantum–Celestial Analogy.*** *We draw a* purely formal *analogy between quantised levels in atomic systems and the discrete orbital radii generated by the SRHF parameterisation. This analogy is conceptual and exploratory; it does not assume a shared physical mechanism.*

Figure 9 juxtaposes two systems typically considered independent: quantised electron orbits in atomic physics and the harmonic spacing of planetary semi-major axes in the present model. Both descriptions feature indexed radii, enabling a schematic, non-dynamical comparison. In atomic systems, these radii are derived from established quantum dynamics; in the planetary case, they result from algebraic constructions based on $\sqrt{2}$, selected *a posteriori* to reproduce the observed Solar System architecture.

The analogy is summarised in Table 3, which juxtaposes selected concepts in atomic physics with their SRHF counterparts.

The silver ratio $\delta_S = 1 + \sqrt{2}$ functions as a scaling factor for the harmonic sequence. It is important to emphasise that this is an observational parallel: there is no suggestion that δ_S plays the role of a fundamental constant analogous to Planck's constant $\hbar$, or that quantum mechanics operates at planetary scales.

The recurrence of similar algebraic forms across different physical contexts has at times raised broader questions about underlying organising principles. The present work identifies such a pattern and makes it available for future theoretical investigation; developing any unified framework is beyond the empirical scope of this article.

8. Conclusions

The fixed SRHF algebraic prescription, based on $\delta_S = 1 + \sqrt{2}$, provides a compact description of the adopted nine observed Solar System semi-major axes (eight planets and Pluto). With Earth as the 1 AU normalisation anchor, and without counting the hypothetical Harmonia location as an observation, the resulting MAPE is 0.792970% and RMSE is 0.105821 AU.

On the explicitly matched six-classical-planet sample, SRHF yields MAPE 0.8954%, compared with 2.6806% for the specified Titius–Bode prescription and 17.5152% for the ideal Kepler-solids construction. These numbers concern descriptive distance fits under stated assumptions, not historical adjacent-ratio errors, physical stability or statistical significance.

The Silver formula singles out an unobserved theoretical node $a_H^{\mathrm{Silver}} = 2.142135624$ AU. The distinct five-outer-body least-squares optimum is $a_H^{\mathrm{opt}} = 2.143687394$ AU; the π-based reference is $a_H^{\pi} = \pi^{2/3} = 2.145029397$ AU. Their closeness is a mathematical feature of these specified constructions. The residual-based HSI has no interior optimum and does not establish gravitational stability.

The framework is therefore best described at present as a phenomenological hypothesis with explicit equations, reproducible fitting results and proposed but not yet independently tested physical implications. Future studies may examine orbital dynamics, small-body distributions, statistical null models and comparative exoplanetary samples. Those investigations are outside the present numerical and interpretative correction.

Computational Methods

The v3.1 correction uses the frozen rounded semi-major-axis inputs and the fixed algebraic SRHF equations, independently verified by the A0 numerical code (NumPy/SciPy with Decimal checks). The adopted values in AU are Mercury 0.387, Venus 0.722, Earth 1.000, Mars 1.524, Jupiter 5.204, Saturn 9.559, Uranus 19.185, Neptune 30.156 and Pluto 39.482. Harmonia is unobserved. The original study cited the JPL Horizons service (JPL Solar System Dynamics, 2023); the author clarifies that these nine rounded figures were compiled from published reference materials, not extracted directly from a single Horizons query. Their precise source editions, orbital-element definitions and historical reference epochs remain incompletely documented; uniform agreement with one modern J2000 table is not claimed. The Earth anchor is included in the nine-observation score, and Pluto is treated as a dwarf planet.

Fixed-model predictions and the conditional comparison prescriptions are specified in the project audit. The five-body empirical Harmonia fit and the HSI are separate diagnostics from the fixed nine-body score. Eleven corrected scientific panels are generated by the v3.1 scripts; three previously supplied graphics are retained without a claimed matching script. The original v3.0 source remains unchanged in the archival version history.

Conflict of Interest

The author declares no commercial or financial relationships that could be construed as a potential conflict of interest.

Author Contributions

Salah-Eddin Gherbi conceived the harmonic framework, developed the mathematical model, implemented the numerical analysis, authored the manuscript and the original TikZ diagrams, and assembled and reviewed the figures. The corrected numerical figures were produced by the identified scripts; Figure 1 was prepared with ChatGPT assistance according to the author's recollection.

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The author acknowledges historical inspiration from Johannes Kepler and the wider tradition of harmonic and geometric approaches to cosmic structure, and thanks early readers for helpful comments on clarity and presentation. The geometric foundations of the Silver-Ratio Harmonic Framework, including the Celtic Cross construction and its relationship to the constants (A, B, C, D), are developed in fuller detail in the author's accompanying book (Gherbi, 2026). Figures 8 and 9 were originally constructed by the author in LaTeX/TikZ for that book; the matching TikZ sources are not included among the manuscript figure generators. The author recalls ChatGPT assistance in preparing Figure 1; the generation record has not been independently checked. The scientific geometry and mathematical interpretation are the author's responsibility.

Data Availability

The v3.1 calculations use nine author-adopted rounded orbital semi-major axes compiled from earlier published references; the exact source editions, element conventions and historical epochs are not fully traceable to a uniform ephemeris extraction. Identified scripts generate eleven corrected scientific figure panels; three supplied graphics have no matched generator in this manuscript repository. The previously released v3.0 materials can be accessed from the public project repository:

`https://github.com/salahealer9/harmonic-architecture-solar-system`

At the time of preparing this revision, corrected v3.1 sources and the independent A2 audit were maintained in the author's signed release-preparation checkout; the public project URL above did not yet expose this exact version. The A2 audit completed with declared limitations. The version-specific manuscript DOI is `10.5281/zenodo.23301593`. Availability of the source package and corrected figures is governed by their own repository or software-release records and should not be inferred solely from the manuscript DOI.

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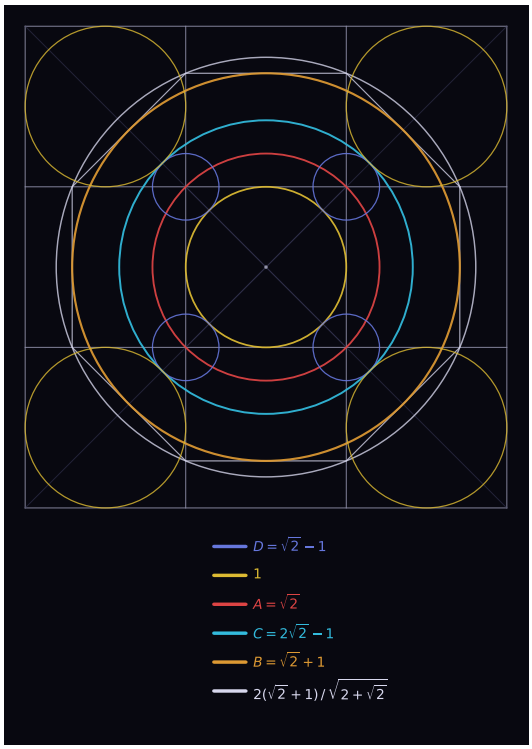


Figure 1. Geometric diagram of the Silver-Ratio Harmonic Framework (SRHF). The 3×3 grid and associated circles illustrate the constants $A = \sqrt{2}$, $B = 1 + \sqrt{2}$, $C = 2\sqrt{2} - 1$ and $D = \sqrt{2} - 1$. The illustration motivates the algebraic parameterisation; it does not derive gravitational dynamics.

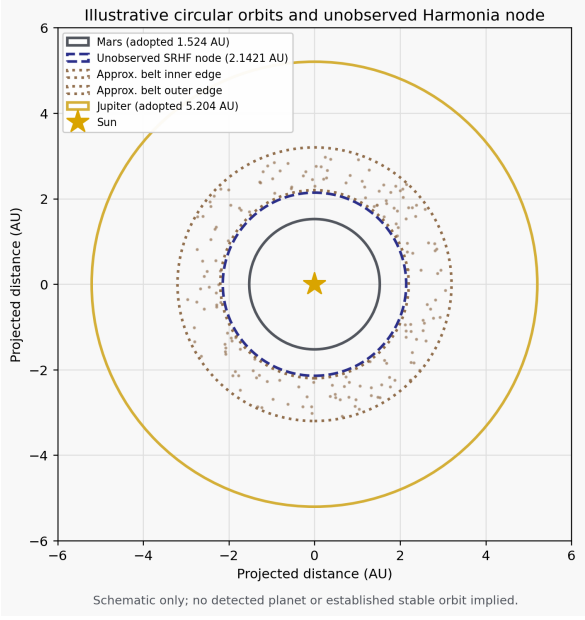
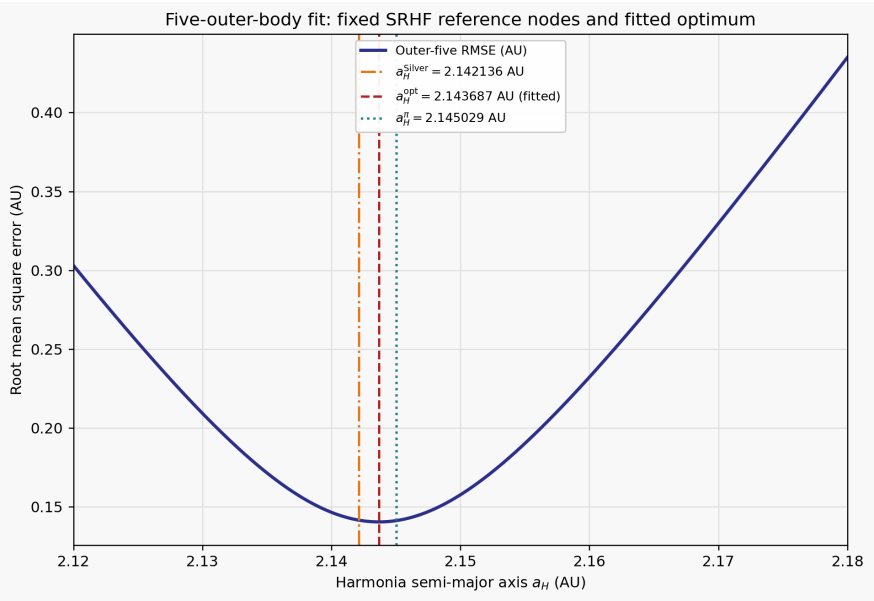
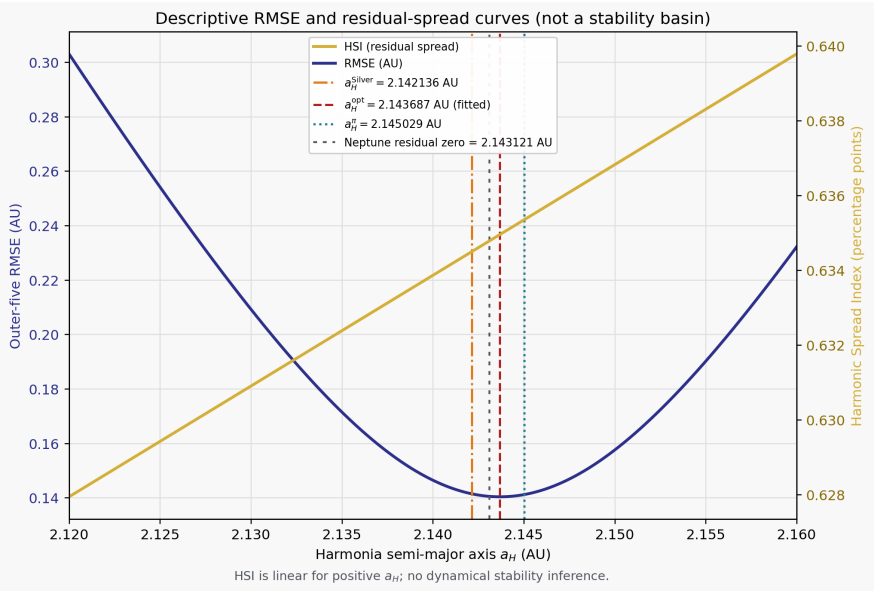


Figure 2. Schematic of the unobserved Silver-Ratio Harmonia location $a_H^{\text{Silver}} \simeq 2.142$ AU between Mars and Jupiter, shown relative to approximate circular orbits and an illustrative asteroid-belt region. Diagrammatic asteroid locations are not an observational density measurement or a detected planet.

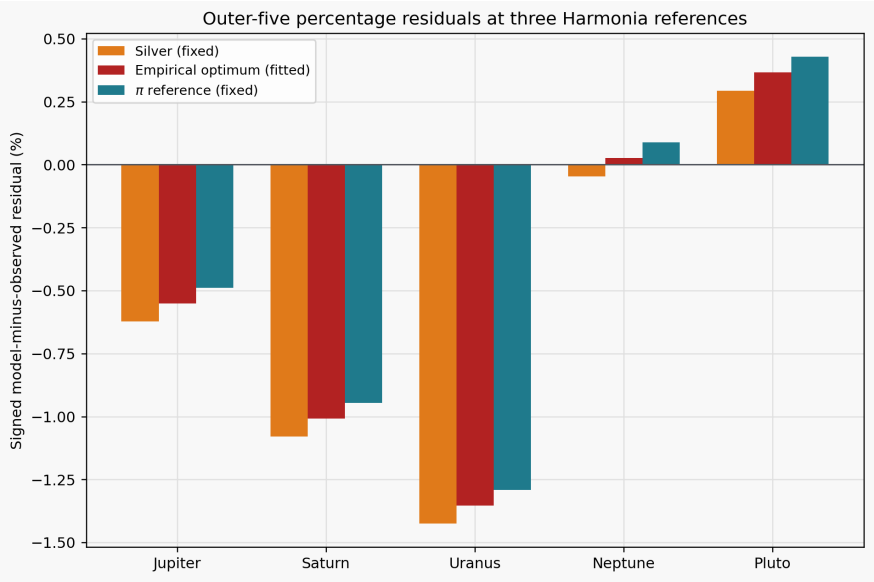


(a) Harmonia optimisation curve.

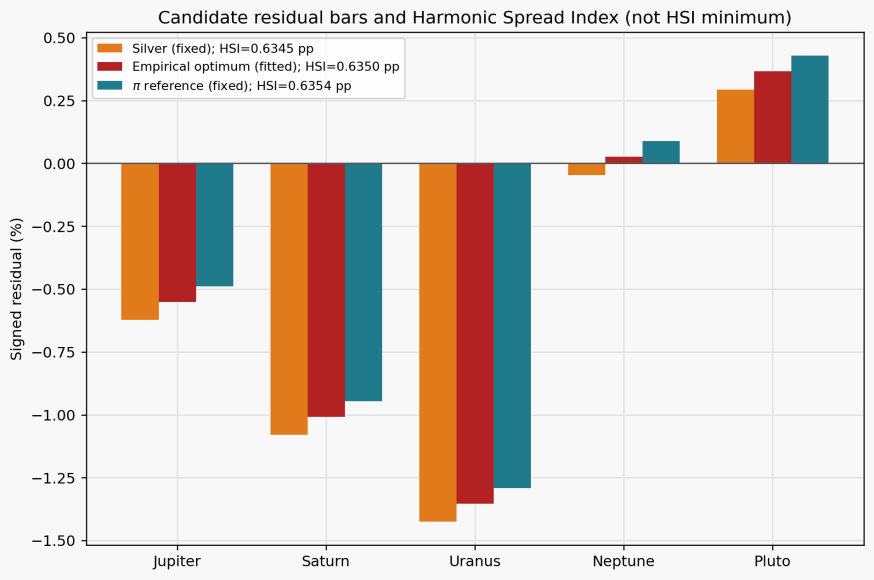


(b) RMSE fit and Harmonic Spread Index.

Figure 3. Harmonia parameter sensitivity for five observed outer bodies. **(A)** The RMSE minimum at the empirically fitted $a_H^{\mathrm{opt}} = 2.143687394$ AU, distinct from the algebraic $a_H^{\mathrm{Silver}} = 2.142135624$ AU and $a_H^{\pi} = \pi^{2/3} = 2.145029397$ AU. **(B)** RMSE (indigo) and Harmonic Spread Index (HSI, gold); the latter is the standard deviation of percentage residuals and is linear in positive a_H. Neither curve by itself demonstrates dynamical stability.

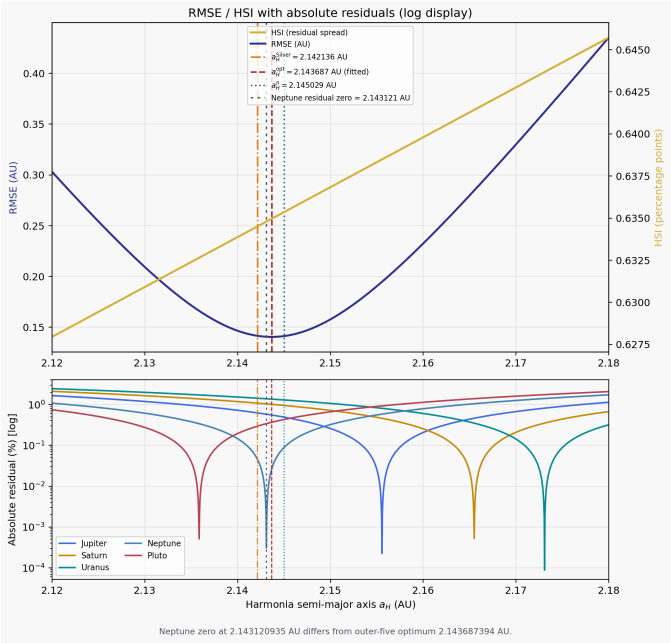


(a) Residuals comparison.

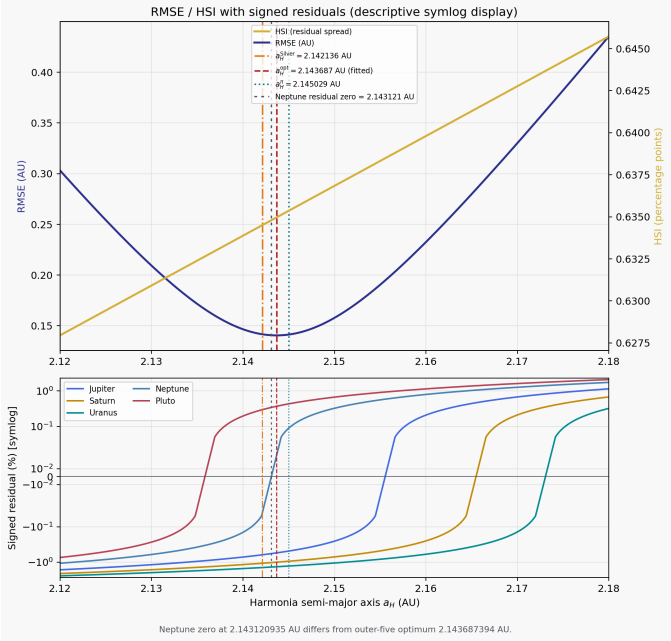


(b) Residual bars with candidate HSI values.

Figure 4. Signed outer-planet residuals for the three specified Harmonia parameter choices. **(A)** Planet-by-planet percentage residuals for a_H^{Silver}, the outer-five fitted a_H^{opt}, and a_H^{π}. **(B)** Grouped residual bars, with the population-standard-deviation HSI reported for each candidate in the legend. Panel B is not an HSI-versus-parameter stability curve.

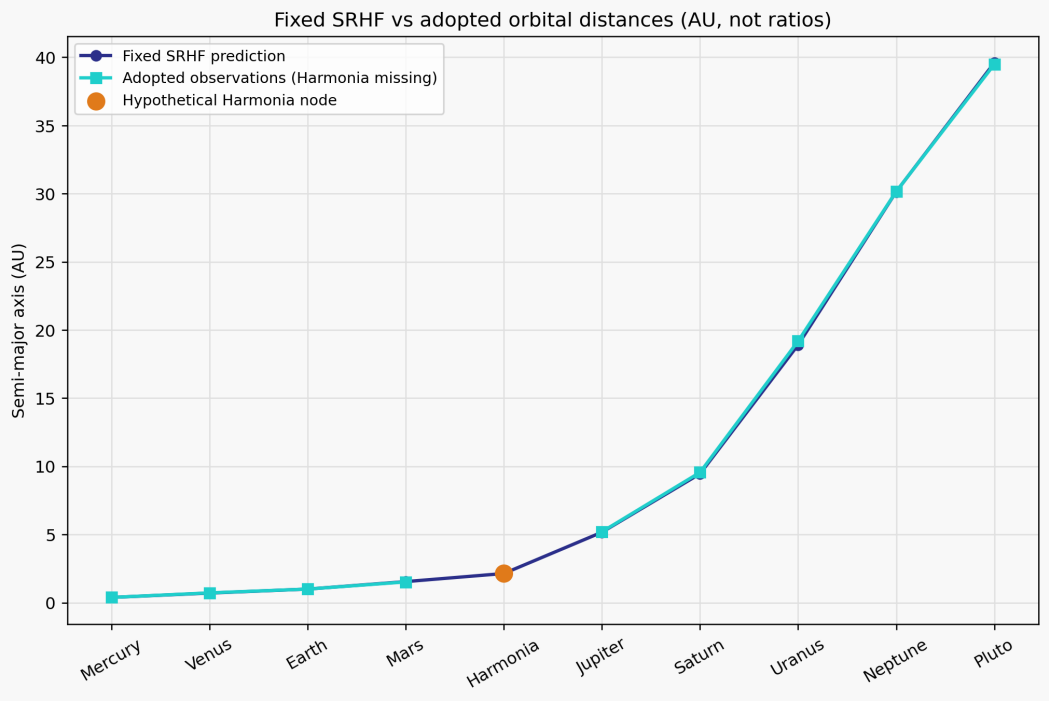


(a) Absolute residual magnitudes on a logarithmic axis.

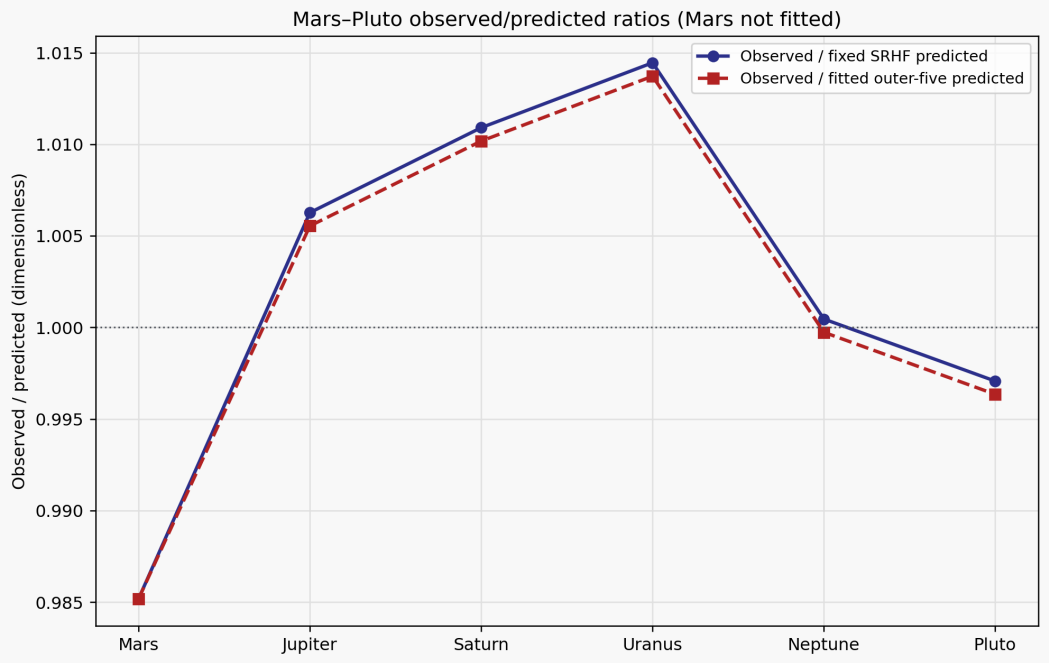


(b) Signed residuals on a symmetric-logarithmic axis.

Figure 5. Outer-planet diagnostic residuals versus the trial Harmonia parameter. **(A)** Absolute percentage residual magnitudes on a logarithmic ordinate. **(B)** Signed percentage residuals on a symmetric-logarithmic scale. Neptuneś residual crosses zero at 2.143120935 AU, distinct from the outer-five RMSE minimum 2.143687394 AU. These transformations do not establish physical resonance or gravitational stability.

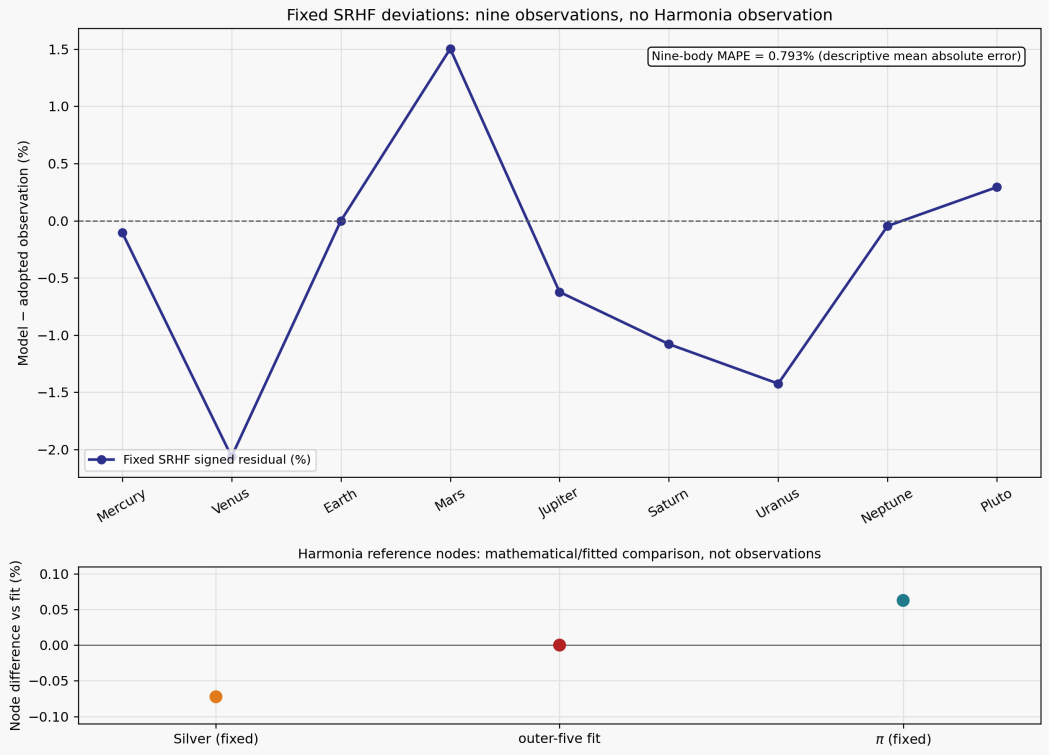


(a) Full-system semi-major axes (AU).

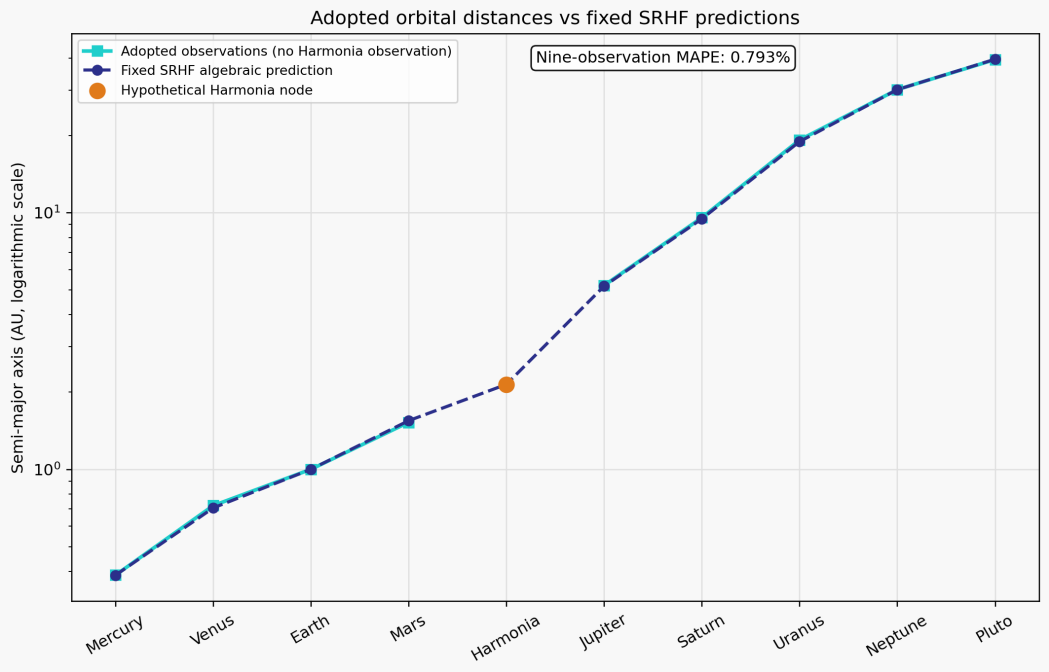


(b) Observed/predicted ratios (Mars–Pluto).

Figure 6. Semi-major-axis comparisons and ratios. **(A)** Observed and fixed SRHF *distances in AU* from Mercury to Pluto; Harmonia is shown solely as a hypothetical prediction. **(B)** Observed/predicted ratios for the six observed bodies Mars, Jupiter, Saturn, Uranus, Neptune and Pluto for the fixed Silver construction and an explicitly fitted outer-body variant. Harmonia is not a measured point in this panel.



(a) Percentage deviations.



(b) Log-scale validation.

Figure 7. Observational residual and distance checks for the fixed algebraic SRHF sequence. **(A)** Percentage residuals for nine observed bodies, with a separate theoretical comparison of the three unobserved Harmonia reference nodes; the nine-observation MAPE is 0.792970% and is not a confidence interval or per-body error bound. **(B)** Observed and fixed-predicted semi-major axes on a logarithmic ordinate, with Harmonia identified only as a model location and excluded from the observational score.

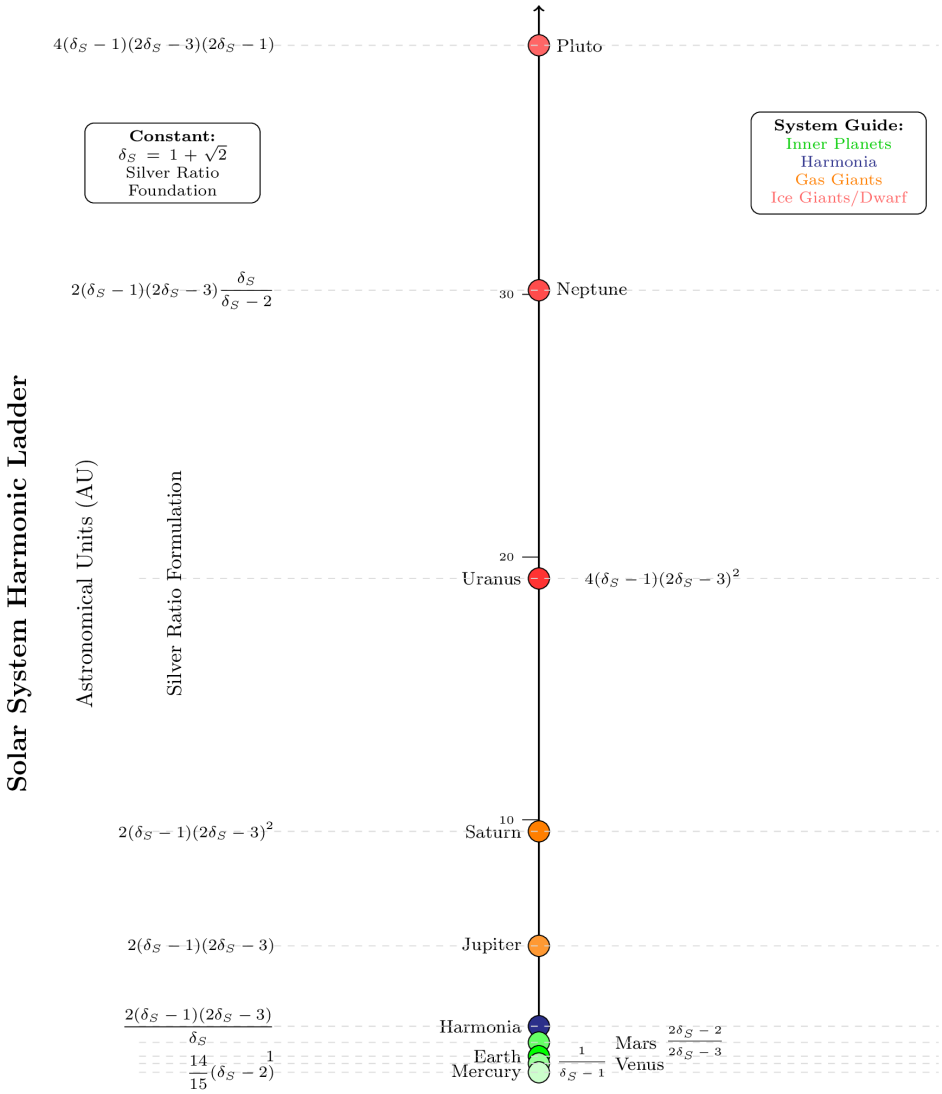


Figure 8. Illustrative Silver-Ratio harmonic ladder: the model assigns successive algebraic radii expressed through $\delta_S = 1 + \sqrt{2}$. These indexed radii are a phenomenological representation, not observed quantum levels or evidence of a planetary-scale quantisation mechanism.

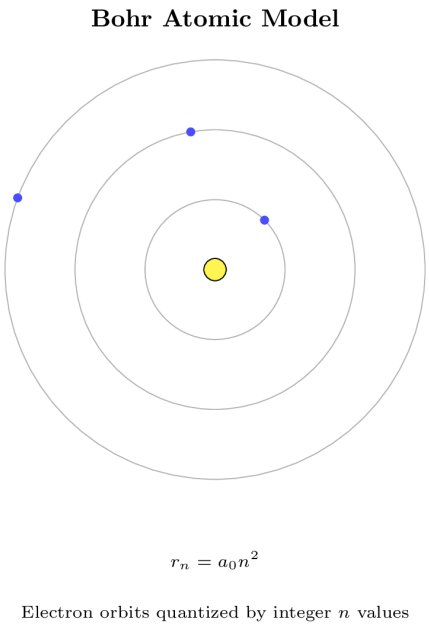


Figure 9. Formal quantum–celestial analogy between indexed atomic-state concepts and discrete entries in the SRHF planetary-spacing parameterisation. The schematic comparison is conceptual only; it does not imply a common physical mechanism or planetary quantum effects.

Table 1. Fixed SRHF model and the adopted rounded semi-major axes. Earth is fixed to 1 AU and included in the score. Pluto is an observed dwarf planet; the hypothetical Harmonia node is unobserved and excluded from all observational errors. Residuals are model minus observation.

Body	Observed (AU)	SRHF (AU)	Δ (AU)	Deviation (%)
Mercury	0.387	0.386599	-0.000401	-0.104
Venus	0.722	0.707107	-0.014893	-2.063
Earth	1.000	1.000000	+0.000000	+0.000
Mars	1.524	1.546918	+0.022918	+1.504
Harmonia*	—	2.142136	—	—
Jupiter	5.204	5.171573	-0.032427	-0.623
Saturn	9.559	9.455844	-0.103156	-1.079
Uranus	19.185	18.911688	-0.273312	-1.425
Neptune	30.156	30.142136	-0.013864	-0.046
Pluto	39.482	39.597980	+0.115980	+0.294
MAPE (9 observed bodies)				**0.792970%**
RMSE (9 observed bodies)			**0.105821 AU**	

** Harmonia* is a theoretical predicted location, not an observation. The historical ten-slot computation, which substituted an assumed Harmonia reference value, is a non-observational diagnostic and is not used in these scores.

Table 2. Matched comparisons under the frozen A0 protocols. The fixed SRHF and comparator predictions are scored with identical observational denominators within each protocol. C6 consists of Mercury, Venus, Earth, Mars, Jupiter and Saturn; C9 additionally includes Uranus, Neptune and Pluto (dwarf planet). Earth is the common 1 AU normalisation anchor and is included in both samples. C9 Titius–Bode is a conditional ordered extrapolation, not a validated physical prediction.

Sample	Model	MAPE (%)	RMSE (AU)
C6 ($N = 6$)	Fixed SRHF	**0.8953985**	**0.0455336**
	Fixed Titius–Bode	2.6805764	0.1829964
	Ideal Kepler solids	17.5152364	1.3689093
C9 ($N = 9$)	Fixed SRHF	**0.7929702**	**0.1058210**
	Titius–Bode (conditional extrapolation)	15.8270030	12.9002128
	Kepler solids	Not applicable	

These semi-major-axis RMSEs are in AU. The older adjacent-orbit-ratio experiment (Kepler MAPE 10.3161%, RMSE 0.239741 dimensionless) is a *different* comparison and must not be substituted for the C6 Kepler entry. The historical Celtic ratio sequence also differs from ratios implied by the fixed SRHF radii.

Table 3. Illustrative structural analogy between selected concepts in atomic physics and in the SRHF parameterisation of planetary spacing. The correspondence is formal and phenomenological only, and is not used in the derivation or quantitative assessment of the model.

Quantum Systems (Atomic Scale)	**SRHF (Planetary Scale)**
Discrete electron orbitals	Discrete harmonic orbital radii in the parameterisation
Quantum numbers (n, l, m)	Integer indices labelling levels of the δ_S ladder
Energy-level spacings	Spacing of modelled semi-major axes from rational functions of δ_S
Radial wavefunction structure	Radial pattern of modelled orbital radii
Stationary bound states	Indexed planetary radii (phenomenological; no stability claim)

Note: The analogies listed above are structural rather than physical. They are intended to assist conceptual interpretation and do not imply quantum effects or microscopic mechanisms operating at planetary scales.